ICMP Smurf Attack Design

CSE 406: Computer Security Sessional

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What is a Smurf Attack?

- Smurf attack using ICMP Echo Requests.
- Attacker spoofs victim's IP and sends to broadcast address.
- Subnet hosts reply to the victim, overwhelming it.

How It Works

1. The attacker sends an ICMP Echo Request to the broadcast address.
2. Spoof victim's IP as source.
3. All amplifiers reply to the victim, causing amplification.

Topology Diagram

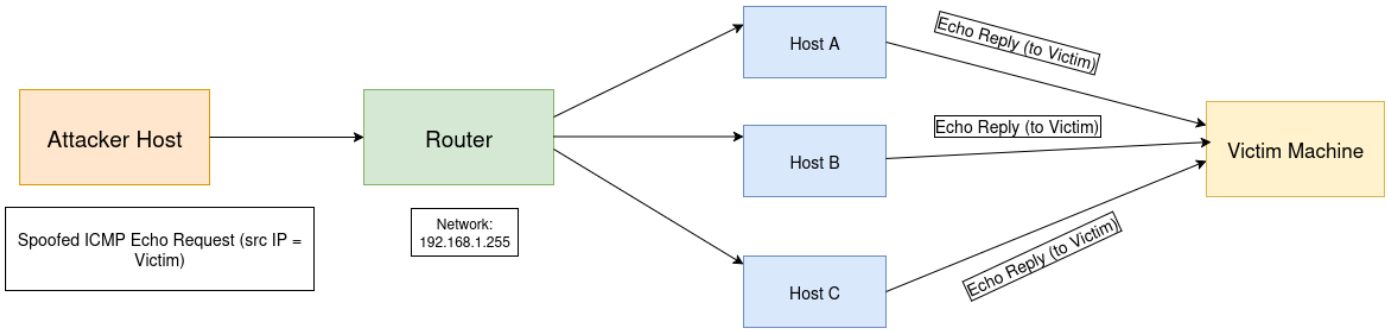
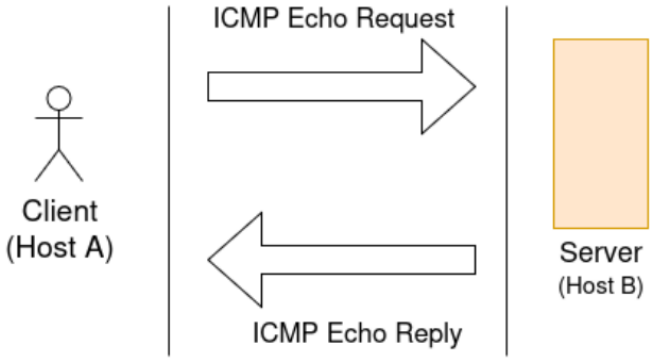


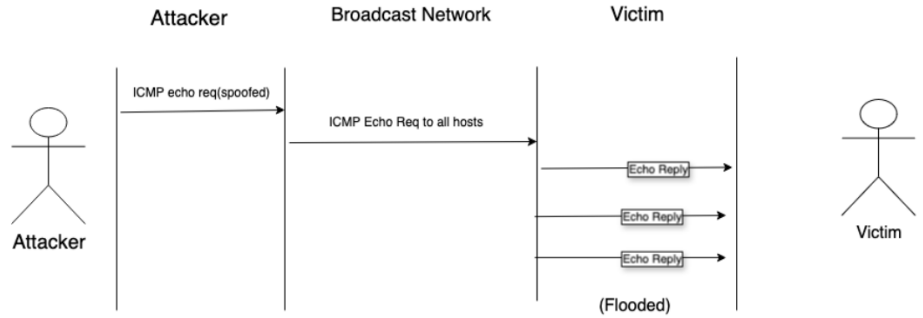
Fig: Topology diagram for ICMP smurf attack

Timing Diagrams

Original Protocol



Attack Timing



Attack Strategy

- Spoof victim's IP in ICMP packet.
- Send to broadcast to get multiple replies.
- Flood victim with traffic.

Packet Details - IP Header

- Source IP: Victim's IP (spoofed)
- Destination IP: Broadcast IP
- Header Checksum: Calculated

Packet Details - ICMP Echo Request

- Type: 8 (Echo Request)
- Code: 0
- Payload includes timestamp and signature.
- The checksum is recalculated.

ICMP Echo Reply - Amplification

- All subnet hosts reply to victim IP.
- The amplification factor will be varied depending on how many amplifiers have been used.
- Victim faces congestion and service disruption.

Why This Design Works

- **Amplification:** 1 request = many replies.
- **Spoofing:** Hides attacker identity.
- **Protocol abuse:** ICMP + broadcast.
- **Efficient:** Low cost, high impact.

Steps of the Attack (Overview)

- Initialize attacker with victim IP, broadcast IP, packet rate.
- Create raw socket with IPPROTO_RAW and broadcast permissions.
- Craft custom IP + ICMP headers (spoofed victim IP).
- Build payload with timestamp, signature, padding.
- Assemble and send packets to broadcast address.
- Track transmission logs, stats, and duration.

Steps of the Attack (Details)

- **Raw Socket:** Created with custom IP headers and broadcast enabled.
- **Crafting:** IP header spoofed with victim IP; ICMP Echo Request built with payload.
- **Spoofing:** Victim's IP used as source to redirect replies.
- **Amplification:** Broadcast causes all hosts to reply to victim.
- **Loop:** Packets sent continuously; logs transmission stats.

Was the Attack Successful?

- **Yes in simulation:**
  - Spoofed packets sent successfully.
  - Hosts replied to victim IP.
  - Logs confirmed continuous sending.
- **In real-world:**
  - Firewalls may block ICMP/broadcast.
  - Routers may disable directed broadcasts.

Observed Output: Attacker Machine

- Logs of each spoofed ICMP Echo Request.
- Packet count and timestamps displayed in real-time.

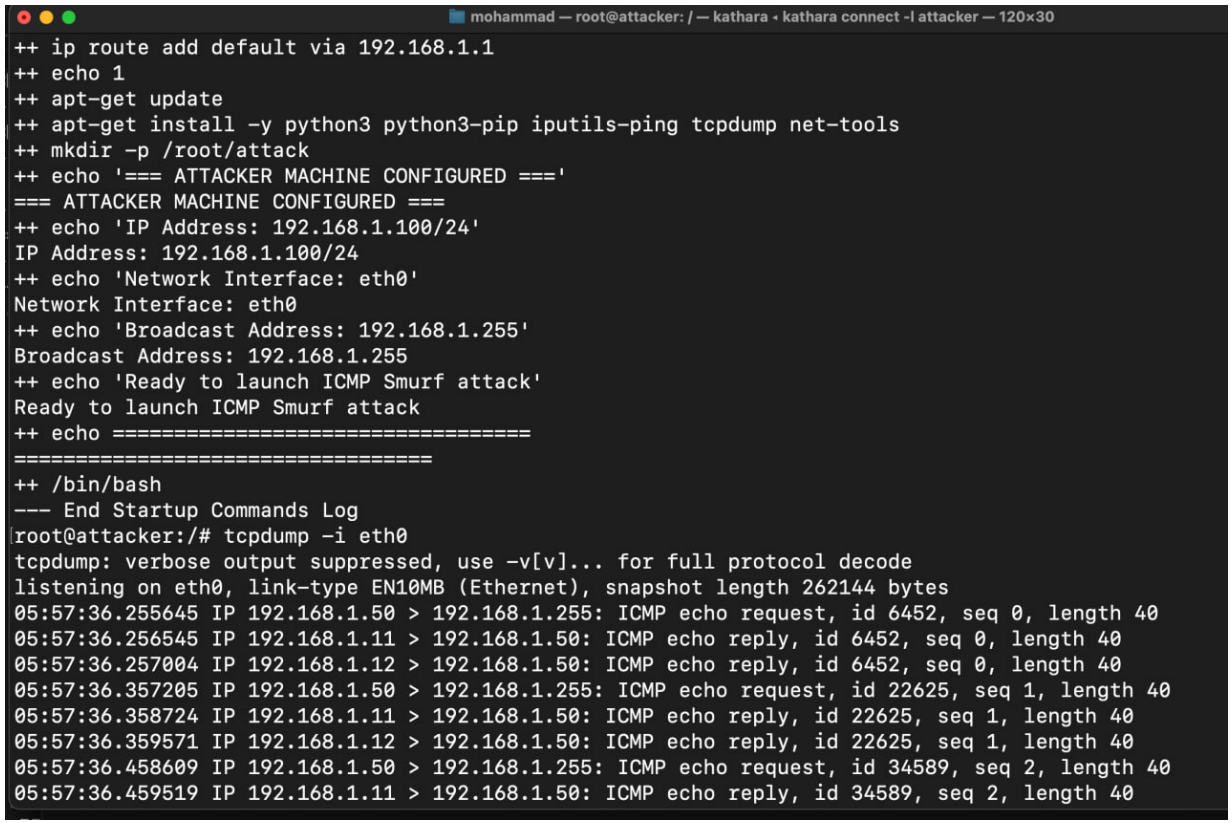


Fig: Attacker terminal log showing packet transmission

Observed Output: Victim Machine

- Received a flood of ICMP Echo Replies.
- Network interface experienced high traffic load.

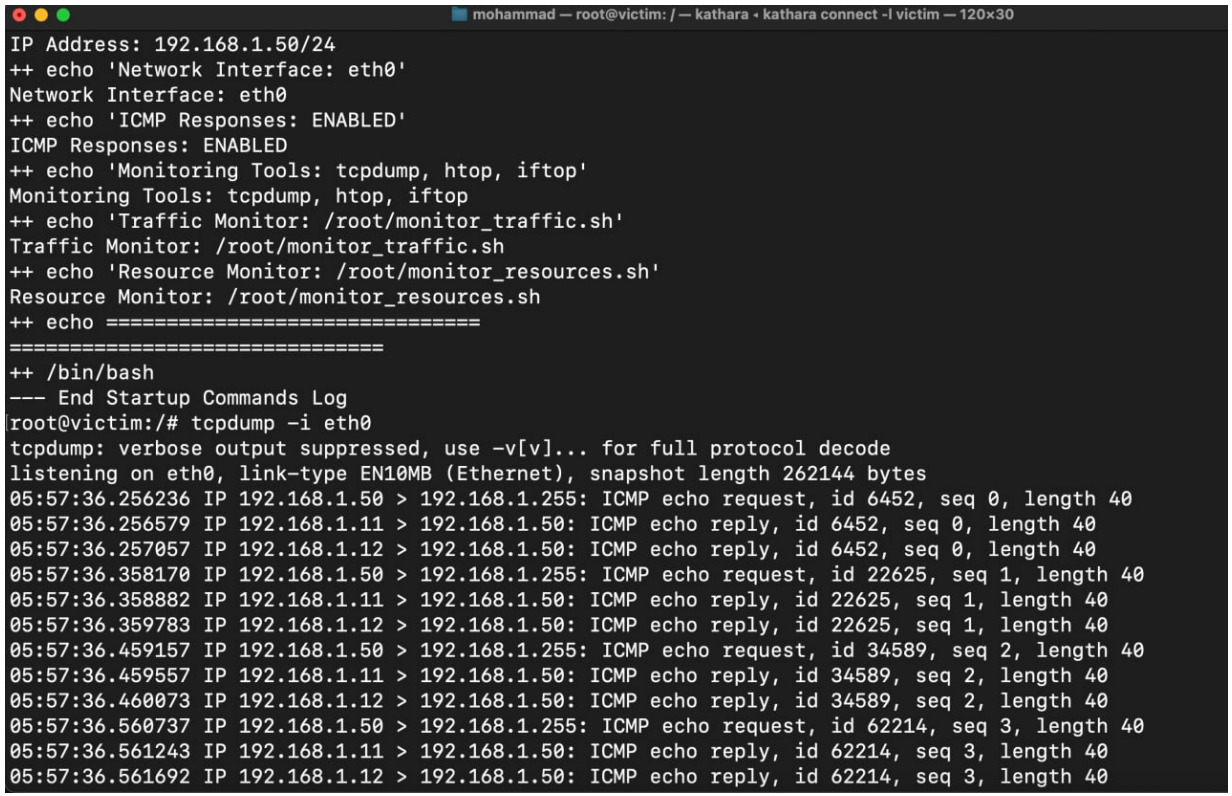


Fig: Victim's network receiving ICMP replies

Observed Output: Other Hosts

- Received ICMP Echo Requests.
- Replied to spoofed victim IP, amplifying the attack.

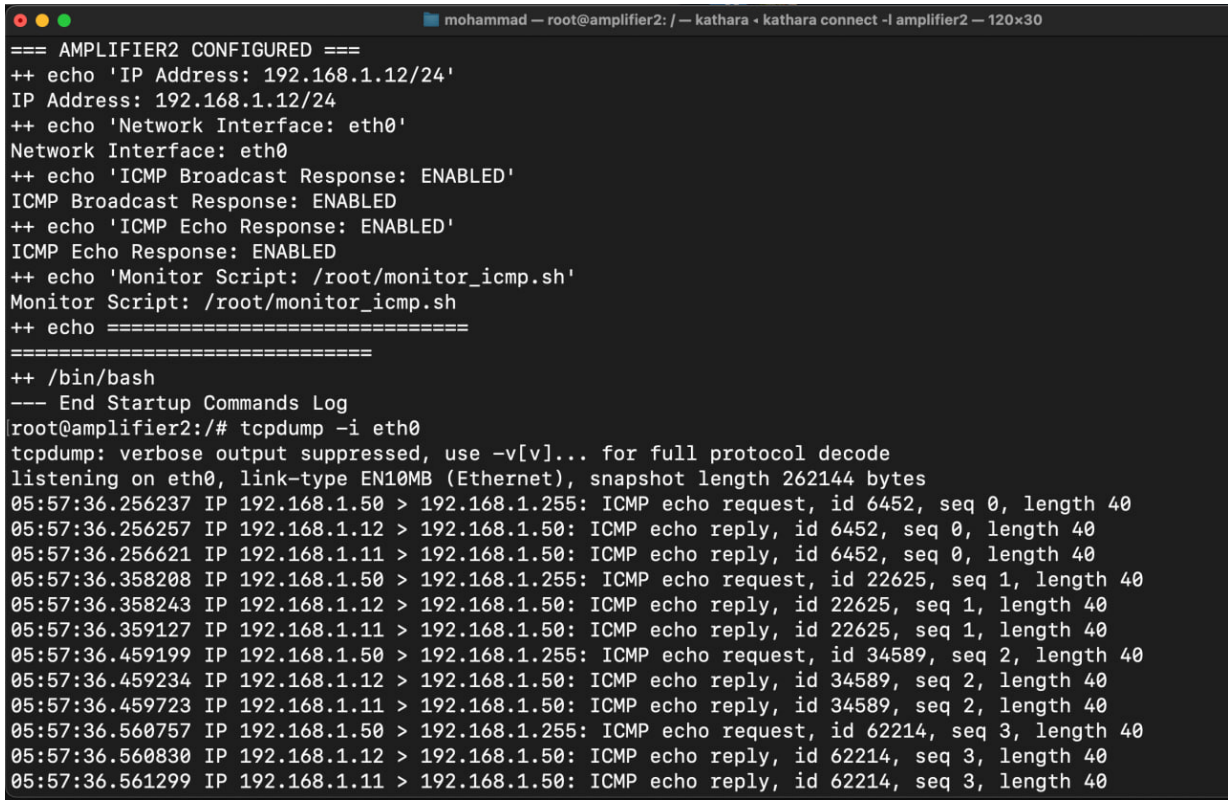


Fig: Other hosts responding to ICMP broadcast

Countermeasures Against Smurf Attacks

- Disable IP-directed broadcasts on routers.

Conclusion

- Smurf attack abuses ICMP and broadcast behavior.
- High amplification potential from a single spoofed request.
- Strong network-level defenses are essential to prevent such attacks.